

Monte Carlo Prediction

Instructor

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Dynamic Programming (DP)



Drawbacks

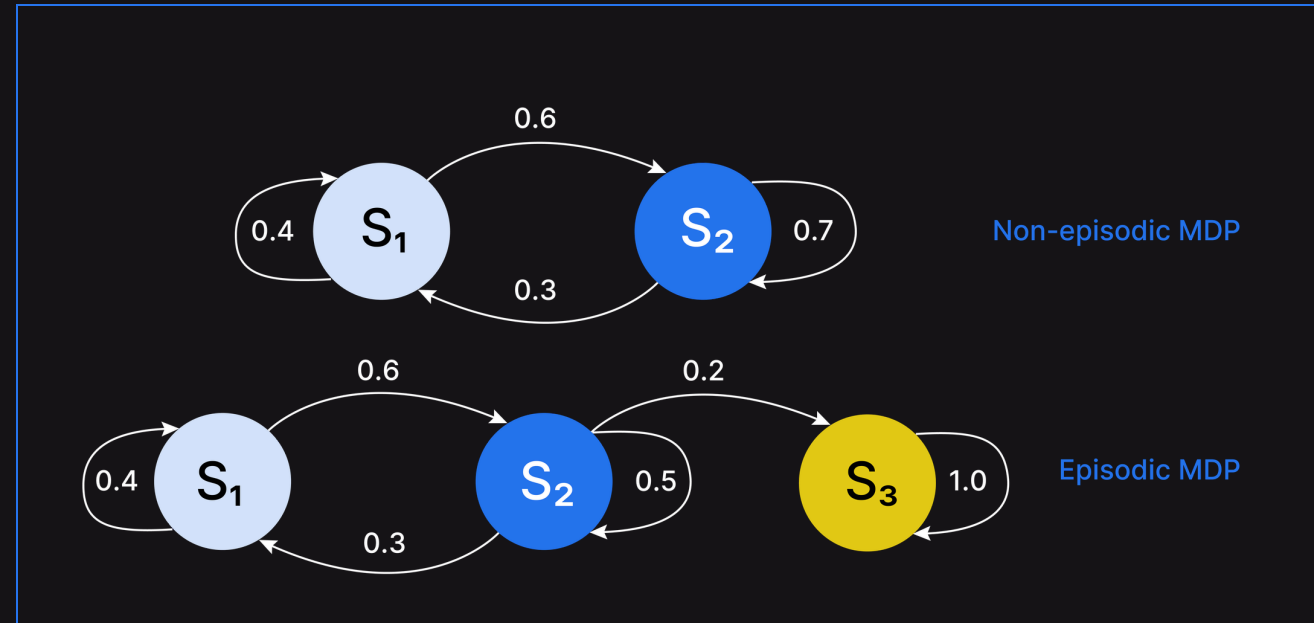
- Requires operations across the entire state-space of the MDP.
- Computationally prohibitive for large state-spaces.
- Assumes full knowledge of MDPs.

Asynchronous DP methods

- States can be updated in any order using available values.
 - Some states may be updated multiple times before others are updated once.

Monte-Carlo (MC) Methods

- MC methods are model-free
- MC methods learn directly from complete episodes of experiences
- Basic idea: Expected return = Empirical mean return
- Episodes must terminate - MC methods apply to episodic MDPs



Monte-Carlo Policy Evaluation

- **Objective:** Given a policy π , estimate V_π from complete episodes of experience

$$S_1, A_1, R_2, S_2, A_2, R_3, \dots, S_T \sim \pi$$

- **Return** is the total discounted reward:

$$G_t = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-t-1} R_T$$

- **Value function** is the expected return

$$v_\pi = \mathbb{E}_\pi[G_t \mid S_t = s]$$



Replace expected return with empirical mean return

First-Visit MC Policy Prediction

- **Objective:** Given a policy π , estimate V_π from complete episodes of experience

Algorithm 1: First-Visit MC Prediction

Input: policy π , positive integer $num_episodes$

Output: value function V ($\approx v_\pi$, if $num_episodes$ is large enough)

Initialize $N(s) = 0$ for all $s \in \mathcal{S}$

Initialize $Returns(s) = 0$ for all $s \in \mathcal{S}$

for episode $e \leftarrow 1$ **to** $e \leftarrow num_episodes$ **do**

 Generate, using π , an episode $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

for time step $t = T - 1$ **to** $t = 0$ (of the episode e) **do**

$G \leftarrow G + R_{t+1}$

if state S_t is **not** in the sequence $S_0, S_1, \dots, S_{t-1}$ **then**

$Returns(S_t) \leftarrow Returns(S_t) + G_t$

$N(S_t) \leftarrow N(S_t) + 1$

end

end

$V(s) \leftarrow \frac{Returns(s)}{N(s)}$ for all $s \in \mathcal{S}$

return V

Every-Visit MC Policy Prediction

- **Objective:** Given a policy π , estimate V_π from complete episodes of experience

Algorithm 2: Every-Visit MC Prediction

Input: policy π , positive integer $num_episodes$

Output: value function V ($\approx v_\pi$, if $num_episodes$ is large enough)

Initialize $N(s) = 0$ for all $s \in \mathcal{S}$

Initialize $Returns(s) = 0$ for all $s \in \mathcal{S}$

for episode $e \leftarrow 1$ **to** $e \leftarrow num_episodes$ **do**

 Generate, using π , an episode $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

for time step $t = T - 1$ **to** $t = 0$ (of the episode e) **do**

$G \leftarrow G + R_{t+1}$

$Returns(S_t) \leftarrow Returns(S_t) + G_t$

$N(S_t) \leftarrow N(S_t) + 1$

end

end

$V(s) \leftarrow \frac{Returns(s)}{N(s)}$ for all $s \in \mathcal{S}$

return V

First-Visit vs Every-Visit

- Consider an MDP with three states A, B and Terminal
- Let us consider two sample runs (episodes):

$$\begin{array}{l} A \xrightarrow{+3} A \xrightarrow{+2} B \xrightarrow{-4} A \xrightarrow{+4} B \xrightarrow{-3} \text{Terminal} \\ B \xrightarrow{-2} A \xrightarrow{+3} B \xrightarrow{-3} \text{Terminal} \end{array}$$

First-Visit

$A \rightarrow^{+3} A \rightarrow^{+2} B \rightarrow^{-4} A \rightarrow^{+4} B \rightarrow^{-3} \text{Terminal}$

$B \rightarrow^{-2} A \rightarrow^{+3} B \rightarrow^{-3} \text{Terminal}$

- For state A

- In first episode, return from the first-visit: +2
- In second episode, return from the first-visit: 0
- $N(A) = 2$
- $\text{Returns}(A) = 2$

$$v(A) = \frac{2}{2} = 1$$

- For state B

- In first episode, return from the first-visit: -3
- In second episode, return from the first-visit: -2
- $N(B) = 2$
- $\text{Returns}(B) = -5$

$$v(B) = \frac{-5}{2} = -2.5$$

Every-Visit

$A \rightarrow^{+3} A \rightarrow^{+2} B \rightarrow^{-4} A \rightarrow^{+4} B \rightarrow^{-3}$ Terminal

$B \rightarrow^{-2} A \rightarrow^{+3} B \rightarrow^{-3}$ Terminal

- For state A

- In first episode, return from every-visit: $2 + (-1) + 1 = 2$
- In second episode, return from the every-visit: 0
- $N(A) = 4$
- $\text{Returns}(A) = 2$

$$v(A) = \frac{2}{4} = 0.5$$

- For state B

- In first episode, return from the first-visit: $(-3) + (-3) = -6$
- In second episode, return from the first-visit: $(-2) + (-3) = -5$
- $N(B) = 4$
- $\text{Returns}(B) = -11$

$$v(B) = \frac{-11}{4} = -2.75$$

Formula: Incremental Mean

Objective: Compute running means $u_1, u_2, \dots$ of a sequence $x_1, x_2, \dots$ incrementally

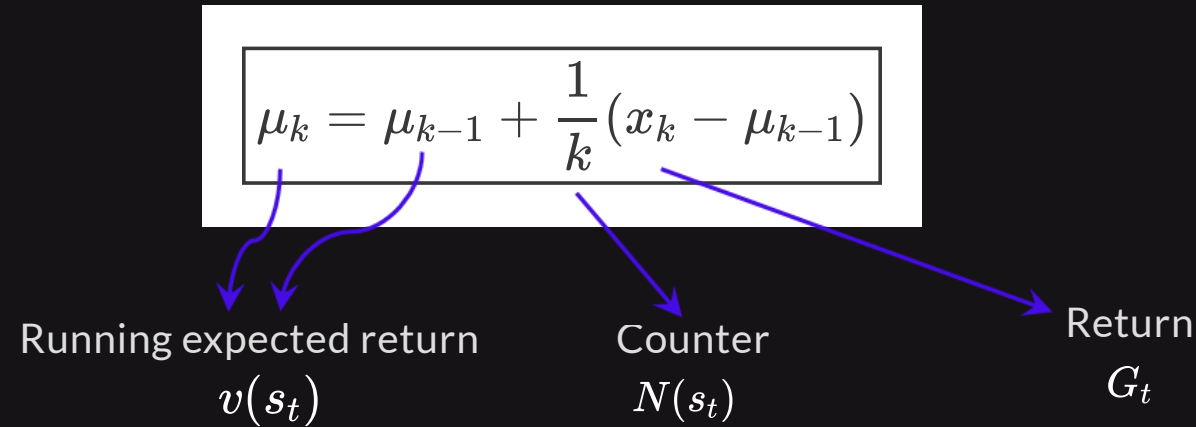
$$\mu_{k-1} = \frac{1}{k-1} \sum_{j=1}^{k-1} x_j$$

$$\mu_k = \frac{1}{k} \left(\sum_{j=1}^{k-1} x_j + x_k \right)$$

$$\mu_k = \frac{1}{k} ((k-1)\mu_{k-1} + x_k)$$

$$\mu_k = \mu_{k-1} + \frac{1}{k} (x_k - \mu_{k-1})$$

Incremental Mean MC Updates



Instead of waiting for all episodes of experiences to get completed, update $v(s)$ incrementally after each episode

For each state s_t with return G_t

$$N(s_t) \leftarrow N(s_t) + 1$$

$$v(s_t) \leftarrow v(s_t) + \frac{1}{N(s_t)}(G_t - v(s_t))$$

Handling Non-stationary Problems

- Forget old episodes
$$v(s_t) \leftarrow v(s_t) + \alpha(G_t - v(s_t))$$

Limitations of Monte Carlo (MC) Methods

- **Dependence on Complete Episodes**

Cannot be used for online learning

- **Low Bias, High Variance**

Sample inefficient due to high variance

- **Lack of Markov Property Utilization**

Inefficient in exploiting the Markov Property

- **Slower Convergence**

Convergence is slower, but guaranteed, even for function approximation

Thank You
